

Chapter 7

Neural Networks and Deep Learning

Learning Outcomes

After completing this chapter, you will be able to:

- Explain the architecture of artificial neural networks
- Implement forward and backward propagation
- Apply activation functions appropriately
- Train neural networks using gradient descent

7.1 Introduction to Neural Networks

Artificial neural networks are computational models inspired by biological neural systems. They consist of interconnected nodes (neurons) organized in layers, each performing simple computations that collectively enable complex pattern recognition and decision-making.

7.1.1 The Perceptron

The simplest neural network is the perceptron, introduced by Frank Rosenblatt in 1958. A perceptron takes multiple inputs, applies weights, and produces a single binary output:

Example 7.1: Binary Classification

Consider a perceptron with two inputs x_1 and x_2 , weights $w_1 = 0.5$ and $w_2 = 0.3$, and bias $b = -0.2$. The output is computed as:

$$y = \text{step}(w_1x_1 + w_2x_2 + b)$$

7.2 Multi-Layer Networks

While perceptrons can solve linearly separable problems, many real-world tasks require multi-layer architectures. A typical feedforward neural network consists of:

- **Input layer:** Receives raw data
- **Hidden layers:** Extract features and patterns
- **Output layer:** Produces final predictions

Figure 7.1: Three-layer neural network architecture



7.3 Activation Functions

Activation functions introduce non-linearity into the network, enabling it to learn complex patterns. Common activation functions include:

Function	Formula	Range	Use Case
Sigmoid	$\sigma(x) = 1/(1+e^{-x})$	(0, 1)	Binary classification
Tanh	$\tanh(x)$	(-1, 1)	Hidden layers
ReLU	$\max(0, x)$	$[0, \infty)$	Most hidden layers
Softmax	$e^{x_i} / \sum e^{x_j}$	(0, 1)	Multi-class output

Key Concepts

- **Forward propagation:** Computing outputs from inputs
- **Backpropagation:** Computing gradients for weight updates
- **Gradient descent:** Optimization algorithm for training
- **Learning rate:** Controls step size during optimization

Chapter Summary

Neural networks are powerful models capable of learning complex patterns from data. By stacking multiple layers and using appropriate activation functions, they can approximate virtually any function. The training process uses backpropagation and gradient descent to iteratively improve the network's parameters.

Exercises

1. Implement a single-layer perceptron in Python
2. Explain why linear activation functions are not useful in hidden layers
3. Calculate the number of parameters in a network with layers [784, 128, 64, 10]
4. Research and explain the vanishing gradient problem